

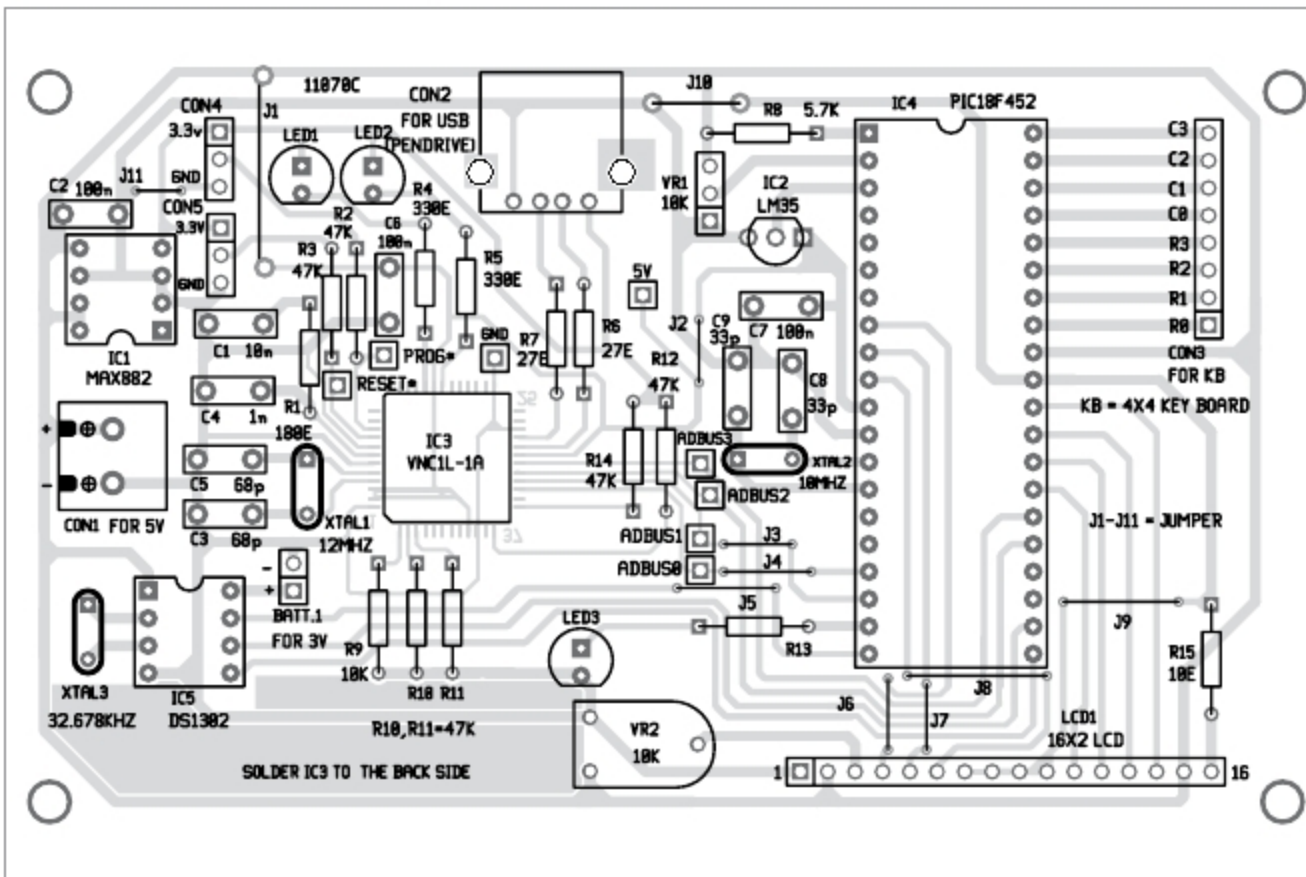


Fig. 5: Components layout for the PCB

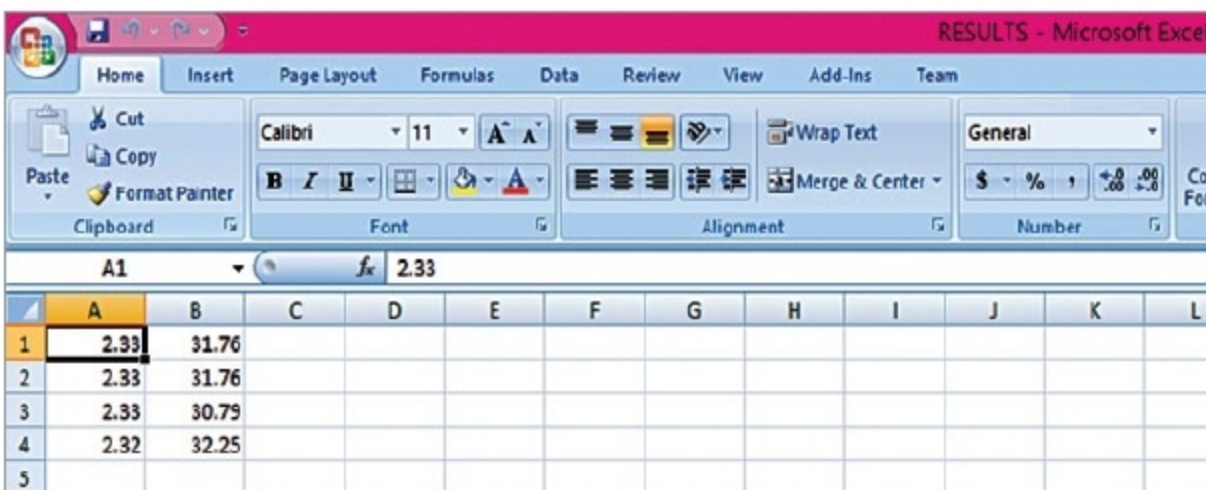


Fig. 6: Result in Excel format

the USB drive—one from LM35 and another through VR1 setting.

Settings and testing

Connect the 4x4 keyboard to CON3 for time settings. An example is given below to feed inputs through the keyboard. Follow the steps given below to test the circuit.

1. Check all circuit connections
2. Connect the circuit board with a 5V, 500mA DC power supply
3. A message reading Starting will be displayed on the LCD. Wait for some time till Press 1 to Start message is displayed on LCD1
4. Press 1 on the keyboard
5. Press 19 for year
6. Press 09 for month
7. Press 08 for date
8. Press 01 weekday (Sunday)
9. Press 08 for hour
10. Press 30 for minute

11. Press 08 for start hour
12. Press 32 start minute
13. Press 08 for end hour
14. Press 35 for end minute
15. Press 00 for hours duration
16. Press 01 for minutes duration

In the example above, data logging starts on October 8, 2019 on Sunday at 8 hours 30 minutes. Actual data logging starts at 8 hours 32 minutes. The process stops at 8 hours 35 minutes. Duration of measurement is within one hour (00 in this case) and with updation cycle of every minute.

Remove the pen drive from the board. Now you can open the recorded data from the USB using Microsoft

Excel or other compatible program. Recorded Data Results file opened using Excel is shown in Fig. 6. The first column shows voltage, while the second shows temperature values. The prototype board was tested with Scandisk 16GB, Sony 32GB and HP 16GB/32GB, to mention a few.

Thus, the USB-based data acquisition system using Vinculum host controller samples the signal at pre-defined intervals of time and stores data in a flash drive.

The, output is saved in Microsoft Excel format, which facilitates quick access to data. Also, Excel format renders further analysis of the data easier. **EFY**

PARTS LIST

Semiconductors:

- IC1 - Max882 voltage regulator
- IC2 - LM35 temperature sensor
- IC3 - VNC1L-1A MCU
- IC4 - PIC18F452 MCU
- IC5 - DS1302 RTC chip
- LCD1 - 16x2 LCD
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 180-ohm
- R2, R3, R10-R12, R14 - 47-kilo-ohm
- R4, R5, R13 - 330-ohm
- R6, R7 - 27-ohm
- R8 - 5.7-kilo-ohm
- R9 - 10-kilo-ohm
- R15 - 10-ohm
- VR1, VR2 - 10-kilo-ohm preset

Capacitors:

- C1 - 10nF ceramic disk
- C2, C6, C7 - 100nF ceramic disk
- C3, C5 - 68pF ceramic disk
- C4 - 1nF ceramic disk
- C8, C9 - 33pF ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - USB A-type connector
- CON3 - 8-pin connector
- CON4, CON5 - 3-pin connectors
- XTAL1 - 12MHz crystal
- XTAL2 - 10MHz crystal
- XTAL3 - 32.678kHz crystal
- Jumper wires
- Pen drive
- PCB

(All these are available from kitsnspares.com)



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EFY Note

The source code of this project is available for free download at source.efymag.com